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Exercise 1.3. A telephone number consists of ten digits, of which the first digit is one of $1, 2, \dots, 9$ and the others can be $0, 1, 2, \dots, 9$. What is the probability that 0 appears at most once in a telephone number, if all the digits are chosen completely at random?

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- 10 digit phone number.
 - 1st digit $\in \{1, \dots, 9\} \rightarrow$ can't be zero.
 - Remaining digits $\in \{0, \dots, 9\}$.
- $\Rightarrow$ Find $P(0 \text{ appears at most once})$

Step 1: Total number of valid phone numbers

- The 1st digit must be $1-9 \rightarrow 9$ options
- Each of the remaining 9 digits can be $0-9 \rightarrow 10$ options per digit

$$\text{Total possible phone numbers} = 9 \cdot 10^9$$

Step 2: Count favorable Outcomes

0 appears at most once (i.e. 0 times or exactly 1 time)

- Case 1: zero appears 0 times
 - 1st digit: 9 options (1-9)
 - Remaining 9 digits: only digits 1-9 (no zero) $\rightarrow 9$ options per digit

$$\text{Number of phone numbers with no zeros} = 9 \cdot 9^9 = 9 \cdot \binom{9}{0} \cdot 9^9$$

- Case 2: Zero Appears Exactly Once
 - 1st digit: 9 options (1-9)
 - Remaining 9 digits: zero must appear in one of the 9 digits. (2-10)
 $\binom{9}{1} = 9$
 Other 8 digits: 9^8

$$\text{Number with exactly one zero} = 9 \cdot 9 \cdot 9^8 = 9^{10} = 9 \cdot \binom{9}{1} \cdot 9^8$$

Step 3: Total Favorable Cases

$$\text{Total Favorable} = (9 \cdot 9^9) + (9 \cdot 9 \cdot 9^8) = 2 \cdot 9^{10}$$

Step 4: Probability

$$P = \frac{\text{Favorable}}{\text{Total}} = \frac{2 \cdot 9^{10}}{9 \cdot 10^9} = \frac{2 \cdot 9^9}{10^9} = 2 \cdot \left(\frac{9}{10}\right)^9$$

$$P = \frac{\text{Favorable}}{\text{Total}} = 0.7748$$

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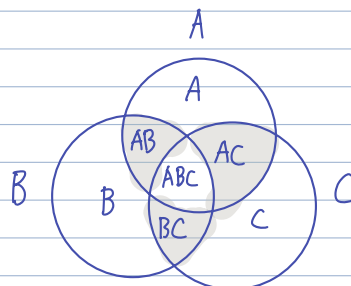
Exercise 1.6 (Skills Exercise). Events A, B , and C are defined in a sample space Ω . Find expressions for the following probabilities in terms of $P(A), P(B), P(C), P(AB), P(AC), P(BC)$, and $P(ABC)$; here AB means $A \cap B$, etc.:

(a) the probability that exactly two of A, B, C occur;

- AB occur, but not C : $P(ABC') \rightarrow A \cap B \cap C'$
- AC occur, but not B : $P(ACB') \rightarrow A \cap B' \cap C$
- BC occur, but not A : $P(BCA') \rightarrow A' \cap B \cap C$

$$P(\text{exactly 2 occur}) = [P(AB) - P(ABC)] + [P(AC) - P(ABC)] + [P(BC) - P(ABC)]$$

$$P(\text{exactly 2 occur}) = P(AB) + P(AC) + P(BC) - 3P(ABC)$$



(b) the probability that exactly one of these events occurs;

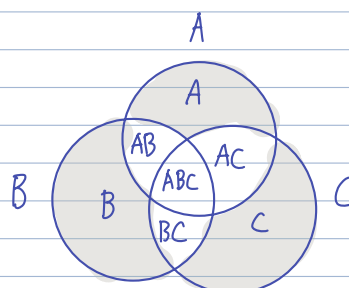
- A only occurs: $P(AB'C') \rightarrow A \cap B' \cap C'$
- B only occurs: $P(BA'C') \rightarrow A' \cap B \cap C'$
- C only occurs: $P(CA'B') \rightarrow A' \cap B' \cap C$

$$P(A \text{ only}) = P(A) - P(AB) - P(AC) + P(ABC)$$

$$P(B \text{ only}) = P(B) - P(AB) - P(BC) + P(ABC)$$

$$P(C \text{ only}) = P(C) - P(AC) - P(BC) + P(ABC)$$

$$P(\text{exactly 1 occurs}) = P(A) + P(B) + P(C) - 2P(AB) - 2P(AC) - 2P(BC) + 3P(ABC)$$



(c) the probability that none of these events occur.

$$P(A' \cap B' \cap C') = 1 - P(\text{At least one of } A, B \text{ or } C)$$

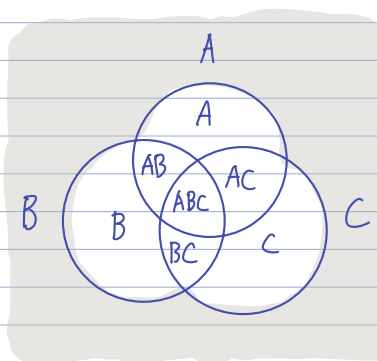
$$P(A \cup B \cup C)' = 1 - P(A \cup B \cup C)$$

$$P(\text{None of } ABC \text{ occur}) = 1 - [P(A \text{ only}) + P(B \text{ only}) + P(C \text{ only})] - [P(AB) + P(BC) + P(AC) - 2P(ABC)]$$

$$= 1 - [P(A) + P(B) + P(C) - 2P(AB) - 2P(AC) - 2P(BC) + 3P(ABC)] - [P(AB) + P(BC) + P(AC) - 2P(ABC)]$$

$$= 1 - P(A) - P(B) - P(C) + \cancel{2P(AB)} + \cancel{2P(AC)} + \cancel{2P(BC)} - \cancel{3P(ABC)} - \cancel{P(AB)} - \cancel{P(BC)} - \cancel{P(AC)} + \cancel{2P(ABC)}$$

$$P(\text{None of } ABC \text{ occur}) = 1 - P(A) - P(B) - P(C) + P(AB) + P(AC) + P(BC) - P(ABC)$$



✓ $A' \cap B' \cap C'$: None of ABC occur

X $A' \cup B' \cup C'$: At least one of ABC did not occur

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Exercise 1.9. In which of the following are events A and B mutually exclusive?

(a)

(a) Roll two dice. A is the event of a sum of 9; B is the event of a double (i.e., the same value on both dice).

$A: (3,6)$ $B: (1,1)$
 $(6,3)$ $(2,2)$
 $(4,5)$ $:$
 $(5,4)$ $(6,6)$

"Can Both Happen?" NO.
 $A \cap B = \emptyset$

No overlapping values, hence A and B are mutually exclusive. ✓

(b) Draw 13 cards from a deck of 52 cards. A is the event of drawing at least one club; B is the event of drawing no aces.

$A: \text{At least one club}$
 $B: \text{No aces}$

"Can Both Happen?" Yes.

Overlapping possibility from drawing Ex: 1 club, 0 aces,
hence A and B are not mutually exclusive.

(c) Toss a coin twice. A is the event of a head on the first toss; B is the event of a head on the second toss.

$A: H$
 $B: H$

"Can Both Happen?" Yes.

Overlapping possibility from flipping both heads,
hence A and B are not mutually exclusive.

④

Exercise 1.13. The letters in the word FULL are rearranged at random. What is the probability that it still spells FULL?

$$\text{Permutation Formula with Repeated} = \frac{n!}{k_1! \cdot k_2!}$$

• $n!$: total # of letters

• $k_i!$: counts of each repeated letter.

Step 1: Total number of letters arrangement

$$\text{Total Permutations} = \frac{4!}{2!} = 12$$

Two repeated L's

• Why not 2? (FUL, L_2, FUL_2L_1)

• Same arrangements in real life, therefore count as one single unique permutations.

Step 2: Count favorable Outcomes = 1

Among the 12 letters arrangements, only one is "FULL".

Step 3: Total Favorable Cases = 1

Step 4: Probability

$$p = \frac{\text{Favorable}}{\text{Total}} = \frac{1}{12}$$

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Exercise 1.16. Twenty cookies are to be distributed to ten children. In how many ways can the cookies be distributed if (a) any cookie can be given to any child; (b) If two cookies are to be given to each child?

◦ Assume cookies and children are distinct.

a)

$$\underset{r=1}{\overset{\text{1st cookie}}{\uparrow}} 10 \cdot \underset{r=2}{\uparrow} 10 \cdot 10 \cdot 10 \dots \underset{r=20}{\uparrow} 10$$

Possible each child gets more than one cookie = 10^{20}
(Not limited by number of cookies)

b) ◦ All Possible Orderings: $20!$

- All unique combinations, order overcounted.
- Double counting pairs. Ex: A/B, B/A

◦ Adjust for indistinguishable groups of 2 cookies / child: $(2!)^{10}$

Each child: $2!$ } $\underset{r=1}{2!} \cdot \underset{r=2}{2!} \cdot \underset{r=3}{2!} \dots \underset{r=10}{2!}$
of groups: 10

◦ Adjust for indistinguishable cookie order overcounted: $10!$
There are $10!$ ways to reorder these groups, should not count.

of ways to form 10 unordered pairs:

$$= \frac{20!}{(2!)^{10} \cdot 10!}$$

internal ordering of each pair ↙
reordering of 10 pairs ↘

(A J)
(B I)
(C H)
(D G)
(E F)
(F E)
(G D)
(H C)
(I B)
(J A)
⋮

Two Cookies for Every Child = $\frac{20!}{(2!)^{10} \cdot 10!} \cdot 10!$ ways to assign 10 pairs to 10 children

Two Cookies for Every Child = $\frac{20!}{(2!)^{10}}$

~~(J A)~~
~~(I B)~~
~~(H C)~~
⋮

With Replacement

Number of ways to choose r objects from n distinct objects if they same subject could be chosen repeatedly:

n^r

Without Replacement

Number of ways to arrange n distinct object with order of arrangement matters:

$n!$

Homework 1

Stat-516

Problem 1.3

Total outcomes = 9×10^9

Possible outcomes = 9^{10} (no zeros) + 9×9^9 (one zero at one place)

probability = $\frac{9^{10} + 9 \times 9^9}{9 \times 10^9} = 0.774841$.

Problem 1.6

Use venn diagram to visualise and get these relations.

a) Exactly two events occur:

A and B occur but not C: $P(AB \setminus (ABC)) = P(AB) - P(ABC)$

similarly other cases can be written.

Adding up all the cases where only two events occur = $P(AB) + P(BC) + P(AC) - 3P(ABC)$

b) Exactly one event occur:

probability exactly A occurs: $P(A) - P(AB) - P(AC) + P(ABC)$

similarly add other cases to get:

Probability only exact one event occurs: $P(A) + P(B) + P(C) - 2P(AB) - 2P(BC) - 2P(AC) + 3P(ABC)$

c) Probability no event occurs = $1 - P(A \cup B \cup C) = 1 - [P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC)]$

Problem 1.9

Mutually exclusive events can't ever occur together by definition. So by definition only option (a) satisfies that criteria. (b),(c) aren't mutually exclusive.

Problem 1.13 The word "FULL" can be rearranged into $\frac{4!}{2!} = 12$ distinct ways. Out of all these rearrangements only one spells out the word "FULL". So the probability is $\frac{1}{12}$.

Problem 1.16

a) Each cookie has 10 (children) options to go to hence no. of ways 10^{20}

b) The first child can have two cookies and these two cookies must be selected out of 20. Thus, the number of choices is $\binom{20}{2}$. Next, two cookies for the next child must be selected out of 18 available; the number of options is $\binom{18}{2}$. Proceeding forward, we have

$$\binom{20}{2} \binom{18}{2} \times \cdots \times \binom{2}{2} = \frac{20!}{2^{10}}$$

options.